**Project Design Phase-I**
**Solution Architecture**

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|---|---|
| Date | 25 October 2026 |
| Team ID | EN25CS3011092 |
| Project Name | Global Olympic Sports Performance Analysis |
| Maximum Marks | 4 Marks |

**Solution Architecture:**
Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

**Solution Architecture Diagram:**



## Component Description Table

| Component Name | Description / Role in Architecture | Technology Used |
|---|---|---|
| Data Source | The four project datasets: athletes.csv (8,000 athlete records with 26 fields), medal_results.csv (910 country results per Games with 19 fields), games_summary.csv (18 Games with 10 fields) and alltime_country_rankings.csv (60 countries with 13 fields). Together they cover 60 countries, 50 sports and the Games from 1992 to 2026. | CSV files |
| Data Connection | Connects Tableau to the CSV files and lets the data be refreshed when a file changes. | Tableau Data Connections (Text File) |
| Data Preparation | Checks data types, nulls and duplicates, relates the files on noc_code and year, assigns geographic roles to the country field and creates a Year field. | Tableau Data Source tab |
| Data Layer | Stores the prepared data model and extract that every worksheet and dashboard reads from. | Tableau data model / extract |
| Analytics / BI Layer | Calculates KPIs and measures such as total medals, medals per athlete, average medal efficiency, sports investment against medals and nations represented, and applies filters, aggregations and the KPI Selector parameter. | Tableau calculated fields and parameters |
| Visualization / Reporting | Presents the insights as charts in two interactive dashboards and a story, with filters and tooltips. | Tableau Desktop (Dashboard 1, Dashboard 2 and Story) |
| Web Front-End | Publishes the dashboards and story and shows them inside a small website. | Tableau Public, Flask (Python), HTML / CSS |
| Business Users | Sports strategists, investment analysts and journalists explore the dashboards to compare countries, sports and regions and to make decisions. | Tableau dashboards (Tableau Public / browser) |